dax_measures.wl

Data Model and DAX Measures

The Power Pivot data model behind the dashboard. The fact table is named `Wildlife`. If your table has another name, replace `Wildlife` in the formulas.

Setup steps

1. **Add the data to the Data Model:** load the cleaned query to the Data Model, or use **Power Pivot > Add to Data Model**.
2. **Rename the table** to `Wildlife`.
3. **Check data types:** `year`, `month` and `day` as Whole Number, `eventDate` as Date, everything else Text.
4. **Add the `Decade` column** (see below).
5. **Add the measures** below by clicking a cell in the grey calculation area under the data grid, typing the formula and pressing Enter.

Calculated column

Decade

Groups years into 1960, 1970, 1980 and so on. It is used to compare classes across decades, which is clearer than 60 separate years.

```
Decade = INT(Wildlife[year] / 10) * 10
```

Note: the 1960 decade covers 1962 to 1969, and the 2020 decade covers 2020 to 2026.

Measures

Why the measures are written this way

- Many columns contain blanks. A plain distinct count would treat a blank as one extra “species”, “genus” or “family”.
- `CALCULATE(DISTINCTCOUNT(...), column <> "")` leaves out blanks, whether they are truly empty or empty text.
- Unmatched taxonomy was labelled `Unknown`, so the phylum, class and order counts also exclude `Unknown`.

Records

The number of occurrence records. It counts every row, including rows with blanks.

```
Records = COUNTROWS(Wildlife)
```

Unique counts

```
Unique Phyla =
CALCULATE(
    DISTINCTCOUNT(Wildlife[phylum]),
    Wildlife[phylum] <> "",
    Wildlife[phylum] <> "Unknown"
)

Unique Classes =
CALCULATE(
    DISTINCTCOUNT(Wildlife[class]),
    Wildlife[class] <> "",
    Wildlife[class] <> "Unknown"
)

Unique Orders =
CALCULATE(
    DISTINCTCOUNT(Wildlife[order]),
    Wildlife[order] <> "",
    Wildlife[order] <> "Unknown"
)

Unique Families =
CALCULATE(
    DISTINCTCOUNT(Wildlife[family]),
    Wildlife[family] <> "",
```

```
    Wildlife[family] <> "Unknown"
)

Unique Genera =
CALCULATE(
    DISTINCTCOUNT(Wildlife[genus]),
    Wildlife[genus] <> ""
)

Unique Species =
CALCULATE(
    DISTINCTCOUNT(Wildlife[species]),
    Wildlife[species] <> ""
)
```

Records with Species

Counts only records that have a species name. It is used for the top-species and rare-species lists, so the blank row does not take first place.

```
Records with Species =
CALCULATE(
    COUNTROWS(Wildlife),
    Wildlife[species] <> ""
)
```

Records per Species

The average number of records per species. It divides by species records only, so records identified to a higher rank do not inflate it. Format it to 1 decimal place.

```
Records per Species =
DIVIDE([Records with Species], [Unique Species])
```

Which measure answers which question

Question	Rows	Columns	Values
Records by class	class		Records

Question	Rows	Columns	Values
Records per year (1962 to 2026)	year		Records
Months with the most sightings	month		Records
Provinces by records and species	stateProvince		Records, Unique Species

Dashboard slicers

Three slicers are connected to every chart through **Report Connections**:

- `Decade`
- `class`